P17: Parametric Equations

1. Introduction to Parametric Form

Instead of relating x and y directly, parametric equations express both coordinates in terms of a third variable called the **parameter**, usually t.

> **Parametric equations:** A curve is defined by $x = f(t)$ and $y = g(t)$, where t ranges over some domain. Each value of t gives a point (x, y) on the curve.

The parameter t often represents time, an angle, or a scaling variable. As t varies, the point $(f(t), g(t))$ traces out the curve.

Why use parametric equations?

- Some curves (e.g. a circle) cannot be expressed as a single function $y = f(x)$
- They naturally describe motion — position at time t
- Easier to differentiate and integrate certain curves

2. Common Parametric Curves

Curve	Parametric form	Cartesian form
Circle (radius r, centre origin)	$x = r\cos t,\ y = r\sin t$	$x^2 + y^2 = r^2$
Circle (centre (a, b), radius r)	$x = a + r\cos t,\ y = b + r\sin t$	$(x-a)^2 + (y-b)^2 = r^2$
Parabola	$x = at^2,\ y = 2at$	$y^2 = 4ax$
Ellipse	$x = a\cos t,\ y = b\sin t$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

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Straight line	$x = x_0 + t\cos\theta,\ y = y_0 + t\sin\theta$	$\frac{y-y_0}{x-x_0} = \tan\theta$

Exam tip: For the parabola $x = at^2, y = 2at$, note the standard result $y^2 = 4ax$. Recognise this family quickly — it appears frequently in exam questions.

3. Converting to Cartesian Form

To eliminate the parameter t and find the Cartesian equation:

1. From one equation, express t in terms of x (or y)
2. Substitute into the other equation
3. Simplify to get a relation between x and y only

Example 1: Find the Cartesian equation of $x = t^2 + 1,\ y = 2t - 3$.

From the second equation: $t = \dfrac{y+3}{2}$
Substitute into the first: $x = \left(\dfrac{y+3}{2}\right)^2 + 1$
So: $x - 1 = \dfrac{(y+3)^2}{4}$, i.e. $(y+3)^2 = 4(x-1)$
This is a parabola with vertex $(1, -3)$.

Example 2: Find the Cartesian equation of $x = 3\cos t,\ y = 2\sin t$.

Use the identity $\cos^2 t + \sin^2 t = 1$:
$\cos t = \dfrac{x}{3}$, $\sin t = \dfrac{y}{2}$
So: $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ — an ellipse.

Key identities for elimination:

- $\sin^2 t + \cos^2 t = 1$ — for trig parametric equations
- $\sec^2 t = 1 + \tan^2 t$ — if $x = a\sec t, y = b\tan t$

- $\cosh^2 t - \sinh^2 t = 1$ — hyperbolic forms (rare)

4. Domain and Range from Parametric Equations

The **domain** of x and the **range** of y are determined by the allowed values of the parameter t.

Example: For $x = \cos t,\ y = \sin 2t$ where $0 \le t \le \pi$:
When $t = 0$: $(1, 0)$. When $t = \pi/2$: $(0, 0)$. When $t = \pi$: $(-1, 0)$.
Domain of x: $[-1, 1]$. Range of y: $[-1, 1]$.
Note: you must trace the curve over the full parameter range to find domain/range correctly.

5. Differentiating Parametric Equations

Chain rule for parametric differentiation:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\dot{y}}{\dot{x}}$$

provided $\dfrac{dx}{dt} \neq 0$.

Second derivative

To find $\dfrac{d^2y}{dx^2}$, differentiate $\dfrac{dy}{dx}$ with respect to x using the chain rule again:

$$\frac{d^2y}{dx^2} = \frac{\dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)}{\dfrac{dx}{dt}}$$

Example: For $x = t^2,\ y = t^3$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

$$\frac{dx}{dt} = 2t,\ \frac{dy}{dt} = 3t^2$$
$$\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2}$$

$$\frac{d}{dt}\left(\frac{3t}{2}\right) = \frac{3}{2}, \text{ so } \frac{d^2y}{dx^2} = \frac{3/2}{2t} = \frac{3}{4t}$$

6. Tangents and Normals to Parametric Curves

At parameter value $t = t_0$:

- **Gradient of tangent**: $m = \left.\frac{dy}{dx}\right|_{t=t_0}$
- **Point on curve**: $(f(t_0),\ g(t_0))$
- **Tangent equation**: $y - g(t_0) = m\big(x - f(t_0)\big)$
- **Normal gradient**: $-1/m$

Example: For $x = t^2,\ y = 2t$, find the equation of the tangent at $t = 3$.

Point: $(9, 6)$. $\frac{dy}{dx} = \frac{2}{2t} = \left.\frac{1}{t}\right|_{t=3} = \frac{1}{3}$
Tangent: $y - 6 = \frac{1}{3}(x - 9)$, so $y = \frac{x}{3} + 3$

Stationary points: occur where $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$. Points where $\frac{dx}{dt} = 0$ give *vertical tangents*.

7. Integrating Parametric Equations

To find the area under a parametric curve between $t = \alpha$ and $t = \beta$:

$$A = \int y\,dx = \int_{\alpha}^{\beta} y(t) \cdot \frac{dx}{dt}\,dt$$

Important: Check whether x is increasing or decreasing as t increases. If decreasing, swap limits or include a negative sign.

Example: Find the area under $x = t^2,\ y = t^3$ from $t = 0$ to $t = 2$.

$$\frac{dx}{dt} = 2t$$
$$A = \int_0^2 t^3 \cdot 2t\,dt = \int_0^2 2t^4\,dt = \left[\frac{2t^5}{5}\right]_0^2 = \frac{64}{5}$$

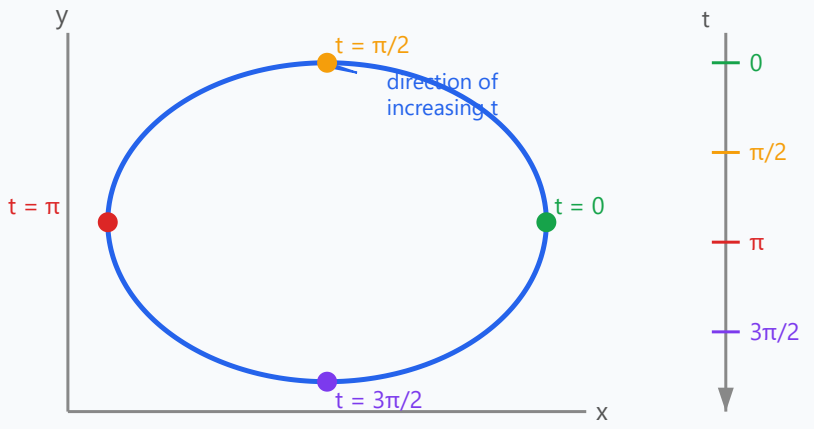
Area enclosed by a closed parametric curve

For a closed curve traced once for $\alpha \le t \le \beta$ (anticlockwise), the enclosed area is:

$$A = \frac{1}{2}\int_{\alpha}^{\beta} \left(x\frac{dy}{dt} - y\frac{dx}{dt}\right) dt$$

8. Visualising Parametric Curves

The diagram below shows how a parameter t moves along an interval and traces a curve in the xy-plane. The arrows indicate the direction of travel as t increases.



x = r cos t, y = r sin t traces a circle as t goes from 0 to 2π

Exam tip — direction matters: When computing a parametric area, always check whether x increases or decreases as t increases. If the curve is traced right-to-left, the integral gives a negative value — take the modulus for area.

9. Summary of Key Formulae

Topic	Formula
First derivative	$\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$
Second derivative	$\dfrac{d^2y}{dx^2} = \dfrac{\frac{d}{dt}(dy/dx)}{dx/dt}$
Area under curve	$A = \displaystyle\int_\alpha^\beta y \frac{dx}{dt}\, dt$
Circle	$x = r\cos t,\ y = r\sin t \Rightarrow x^2 + y^2 = r^2$
Parabola	$x = at^2,\ y = 2at \Rightarrow y^2 = 4ax$